Visualizing data from microplate OD600

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Load and data wrangle

Load and wrangle data

```
#load data, change date
data<-read.csv("../data/20250606OD600.csv")
design<-read.csv("../reference/20250606OD600Design.csv", header=FALSE)

source("C:/Users/parke/OneDrive - Indiana University/GitHub/ASG-fitness-effects/microplate_fitness_assay
```

Prep Data

```
#FEP, Fitness Effects of Phage (subtract Ctrl Infection avgs from corr Strain/Media rows)
# ctrl avgs from od.long
ctrl_avg <- od.long %>%
  filter(Infection == "ctrl") %>%  # Keep only control rows
  group_by(Media, Strain) %>%
  summarize(ctrl_avg = mean(total.OD, na.rm = TRUE), .groups = "drop")
# subtract control average
FEP <- od.long %>%
  left_join(ctrl_avg, by = c("Media", "Strain")) %>%
  mutate(total.OD = ctrl_avg - total.OD) %>%
  select(-ctrl_avg)  # Remove temporary column
FEP<-FEP%>%  #Fitness effects of Phage
  filter(!grepl("ctrl", Infection))


#FES, Fitness effects of Strain (subtract avg WT values from each corr Media/Infection rows)
# wt avgs from od.long
WTavg <- od.long %>%
  filter(Strain == "WT") %>%  # Keep only control rows
  group_by(Media, Infection) %>%
  summarize(WTavg = mean(total.OD, na.rm = TRUE), .groups = "drop")
# subtract control average
FES <- od.long %>%
  left_join(WTavg, by = c("Media", "Infection")) %>%
  mutate(total.OD = total.OD - WTavg) %>%
  select(-WTavg)  # Remove temporary column
FES<-FES%>%  #Fitness effects of Phage
  filter(!grepl("WT", Strain))
```

```r
#Infection vs no infection
FEPa <- aov(total.OD ~  Strain * Infection , data = FEP)
summary(FEPa)
```

```
##                  Df Sum Sq Mean Sq F value   Pr(>F)
## Strain            3  17339    5780  18.328 4.61e-08 ***
## Infection         3   8682    2894   9.177 6.59e-05 ***
## Strain:Infection  9   6684     743   2.355   0.0272 *
## Residuals        48  15137     315
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
#Mutant vs WT
FESa <- aov(total.OD ~  Strain * Infection , data = FES)
summary(FESa)
```

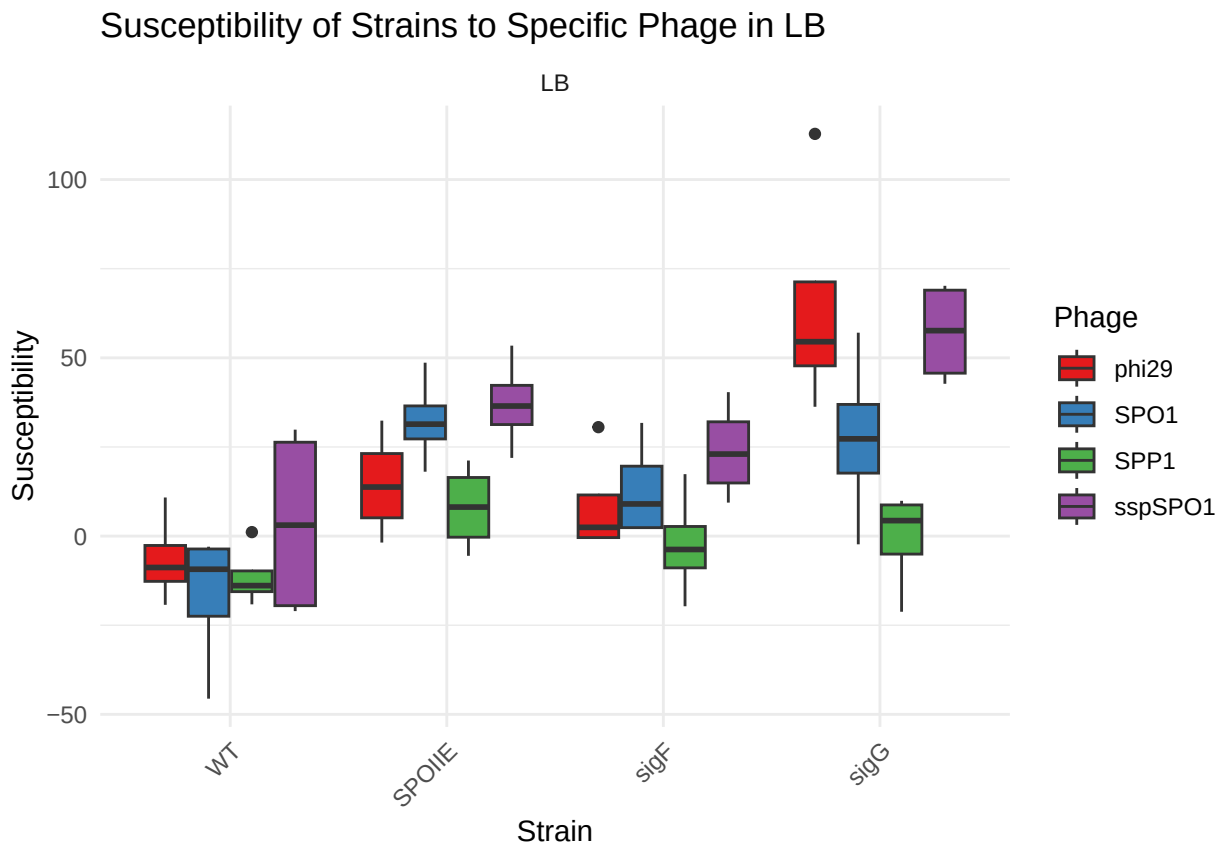
```
##                  Df Sum Sq Mean Sq F value   Pr(>F)
## Strain            2   7434    3717  13.489 1.60e-05 ***
## Infection         4  21318    5330  19.342 4.24e-10 ***
## Strain:Infection  8   7135     892   3.237   0.0042 **
## Residuals        57  15706     276
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Visualize data

```r
# Order strain categories (if needed)
custom_order <- c("WT", "SPOIIE", "sigF", "sigG")  # Adjust as needed

setwd('../Output')

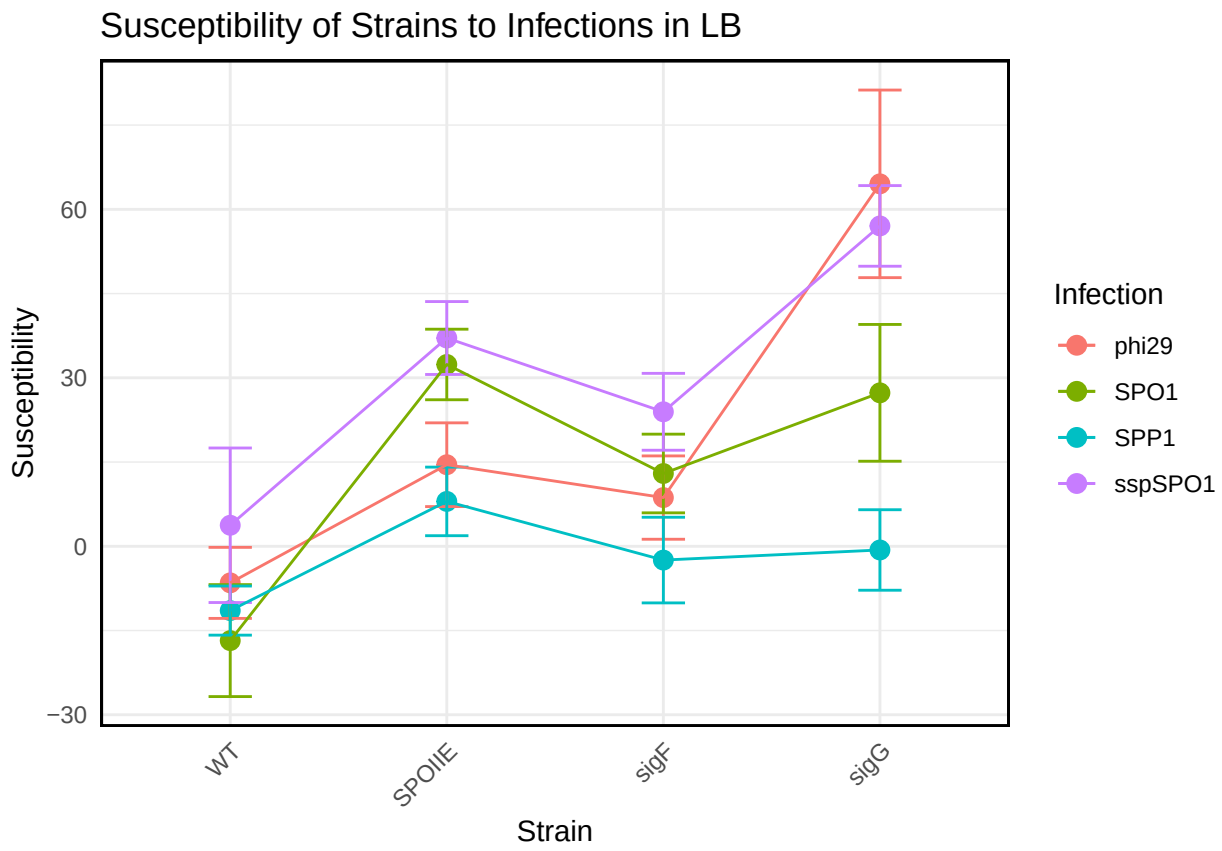

#Box and Whiskers
BWLB<-ggplot(subset(FEP, Media == "LB") %>%
               mutate(Strain = factor(Strain, levels = custom_order)),
             aes(x = Strain, y = total.OD, fill = Infection)) +
  geom_boxplot(aes(group = interaction(Strain, Infection)),
               position = position_dodge(width = 0.8)) +
  theme_minimal() +
  scale_fill_brewer(palette = "Set1") +  # Adjust colors prn
  labs(title="Susceptibility of Strains to Specific Phage in LB",x = "Strain", y = "Susceptibility", fi
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  facet_wrap(~Media)
ggsave("20250606_BWLB.png", plot=BWLB, width = 10, height = 6, dpi = 300)
BWLB
```




```
#Line plot
LPLB<-ggplot(subset(FEP, Media == "LB")%>%
               mutate(Strain = factor(Strain, levels = custom_order)), aes(x = Strain, y = total.OD, co
  stat_summary(fun = mean, geom = "point", size = 3) +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2) +
  stat_summary(fun = mean, geom = "line") +
  theme_minimal() +
  labs(title = "Susceptibility of Strains to Infections in LB",
       x = "Strain",
       y = "Susceptibility") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        panel.border = element_rect(color = "black", fill = NA, size = 1))
```

```
## Warning: The ‘size‘ argument of ‘element_rect()‘ is deprecated as of ggplot2 3.4.0.
## i Please use the ‘linewidth‘ argument instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

```
ggsave("20250606_LPLB.png", plot=LPLB, width = 10, height = 6, dpi = 300)
LPLB
```



SPOIIE was more susceptible to phage in general in comparison to WT. Not much seems to be known about the functions of sigF-activated genes besides that they are expressed in the early forespore while sigG-activated genes are expressed in the late forespore.

SPO1 and sspSPO1 (the removed sspD gene) behave similarly but it seems sspSPO1 is more virulent than SPO1. I'm curious to see if there's a trade off (like having the sspD gene could lead to higher UV or heat resistance)

Plot OD for ref

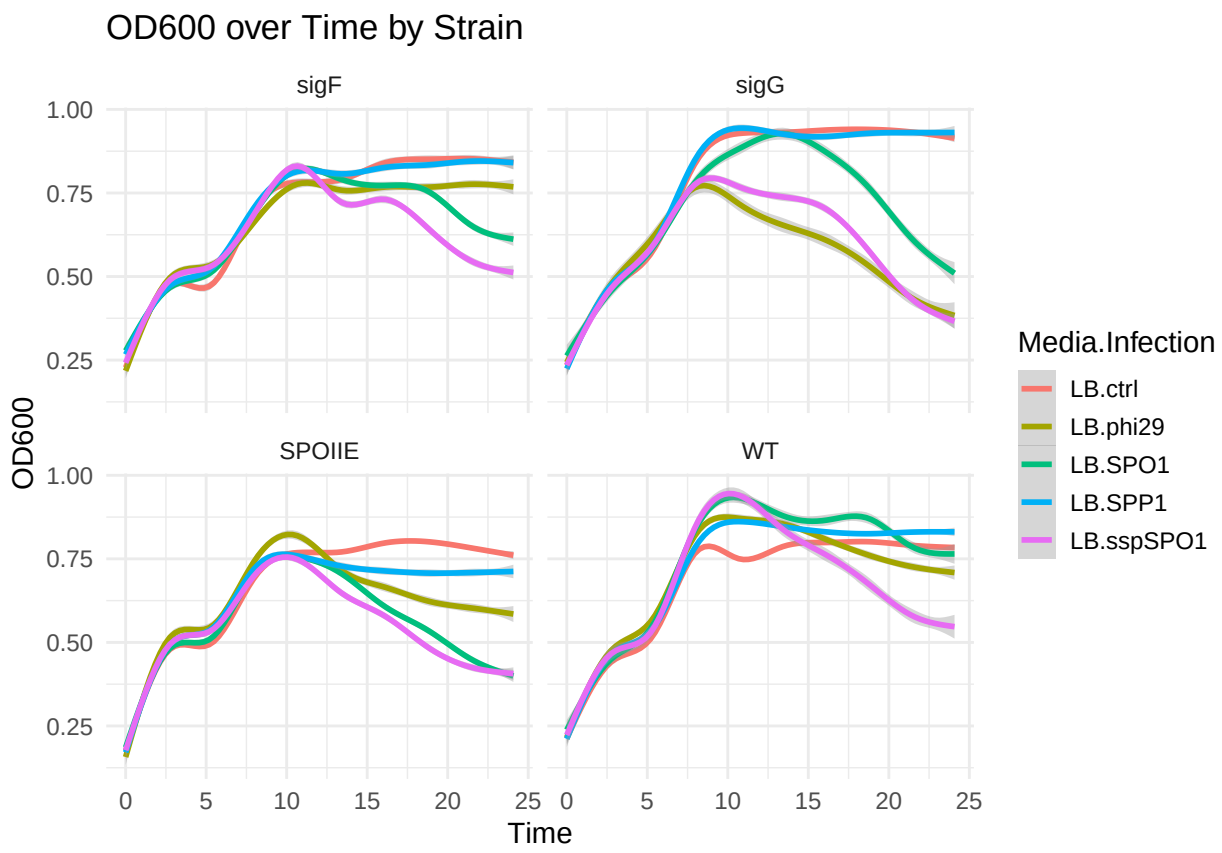
```
setwd('../Output') #for ggsave

#plot and sort by treatment
d<-ggplot(OD.long, aes(x = Time, y = OD600, colour = interaction(Media, Infection))) +
  #geom_line() +
  theme_minimal() + geom_smooth()+
  labs(title = "OD600 over Time by Strain", x = "Time", y = "OD600", colour = "Media.Infection") +
  theme(legend.key.size = unit(0.5, 'cm'), legend.position = "right")+
  facet_wrap(~Strain)
ggsave("20250606_Strains.png", plot = d, width = 6, height = 4, dpi = 300)
```

```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

```
print(d)
```

```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

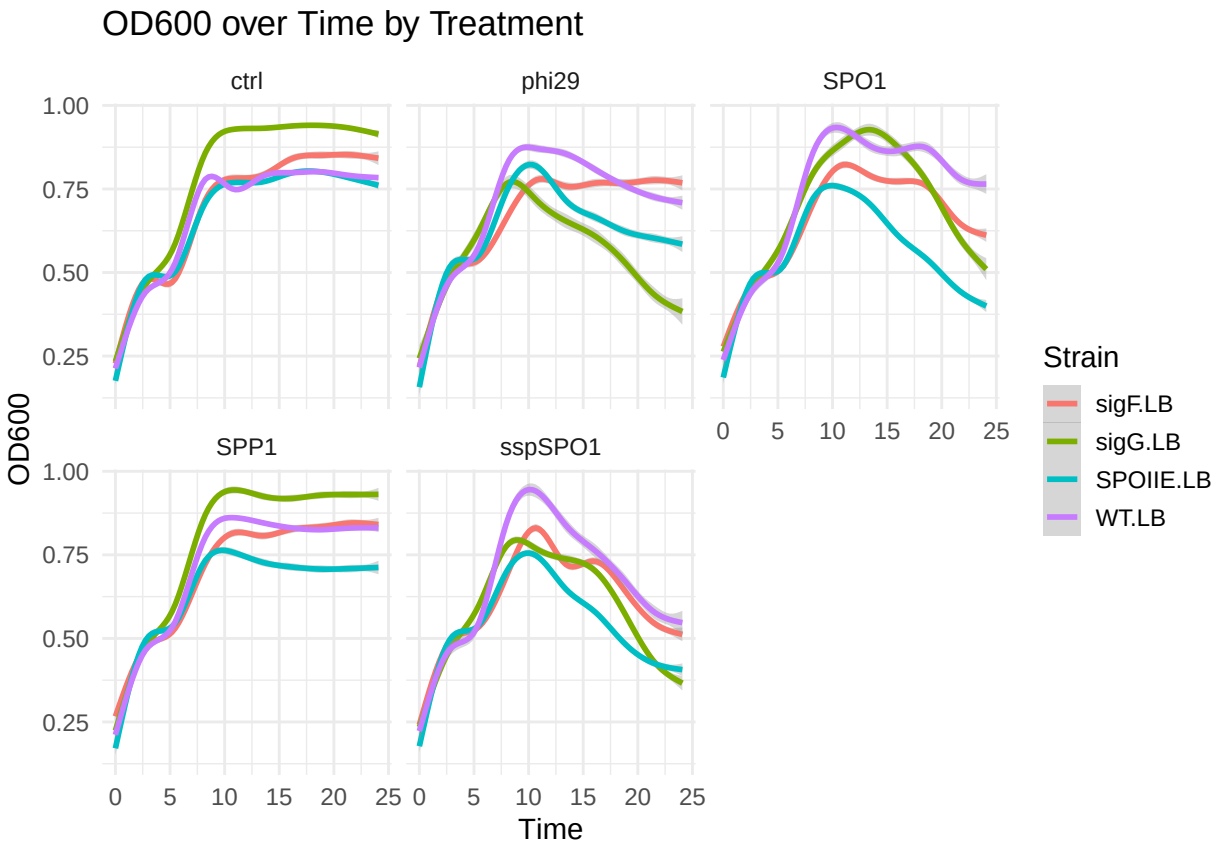



```
#plot and sort by infection
s <- ggplot(OD.long, aes(x = Time, y = OD600, colour = interaction(Strain, Media))) +
  theme_minimal() +
  geom_smooth() +
  labs(title = "OD600 over Time by Treatment", x = "Time", y = "OD600", colour = "Strain") +
  theme(legend.key.size = unit(0.5, 'cm'), legend.position = "right") +
  facet_wrap(~Infection)
ggsave("20250606_Infection.png", plot = s, width = 6, height = 4, dpi = 300)
```

```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

```
print(s)
```

```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```


OD600 over Time by Treatment
ctrl
phi29
SPO1
SPP1
sspSPO1
OD600
1.00
0.75
0.50
0.25
0
5
10
15
20
25
Time
Strain
sigF.LB
sigG.LB
SPOIIE.LB
WT.LB